

YANCHENG (MACY) XIANG

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EDUCATION

University of Wisconsin–Madison

B.S. in Computer Science

GPA: **3.85 / 4.00** — Dean's List: Spring 2022–2023, Fall 2023–2024

Madison, WI

Jan 2023 – Expected May 2026

Skidmore College

Undergraduate Study (Undecided Major)

Saratoga Springs, NY

Sep 2021 – May 2022

INTERNSHIP EXPERIENCES

Hangzhou Xinxiang Infinite Technology Co., Ltd.

AI Product Intern

Hangzhou, China

Jun 2025 – Sep 2025

- Developed and deployed **Python-based SEO automation scripts** integrating Semrush and Google Search Console APIs to track rankings, click-through rates, and coverage; increased organic impressions by **22%** and reduced manual tracking time by 40 hours monthly.
- Designed and implemented a **Flask-based data dashboard** with RESTful endpoints and Matplotlib visualizations to monitor keyword anomalies, CTR fluctuations, and crawl errors in real time, cutting analytics latency by **40%**.
- Created an AI-assisted content workflow using **OpenAI API + Stable Diffusion** that automated prompt generation, post-processing, and A/B testing, reducing design turnaround by **35%** while maintaining brand tone consistency across digital assets.
- Fine-tuned **OCR and NLP models** with spaCy and Tesseract to evaluate text quality within AI-generated outputs, boosting textual accuracy by **18%** and cutting human review effort by 25 hours per week.

Shanghai Yating Biotechnology Co., Ltd. (HLMR Skincare)

Software Engineering Intern

Shanghai, China

Nov 2024 – May 2025

- Engineered a full-stack e-commerce WeChat Mini Program using native frameworks (**WXML, JavaScript**) and a **Node.js** backend, successfully onboarding **200+ registered users** and managing a catalog of **100+ SKUs**.
- Designed and implemented **RESTful APIs** in Node.js to power a dynamic user membership and reward points system, automating point accumulation, tier upgrades, and redemption workflows.
- Optimized front-end rendering performance by implementing component-based architecture and image lazy loading, ensuring smooth scrolling and reducing initial load times for product galleries.
- Integrated the **WeChat Login API** for seamless user authentication and state management, utilizing token-based session controls to securely synchronize shopping cart and user data between client and server.

PROJECT EXPERIENCE

PBS Wisconsin Slack AI Chatbot — *Python, Node.js, RAG, NLP, Slack Bolt, Airtable* Jan 2026 – Present

- Architecting an AI-based Slack chatbot using the Slack Bolt SDK and Airtable API to streamline internal data retrieval for project and staff management.
- Implementing a Retrieval-Augmented Generation (RAG) pipeline with Natural Language Processing (NLP) to interpret user queries and fetch verified data strictly from the Airtable database.
- Designing a comprehensive "Safe Fail-State" mechanism to eliminate LLM hallucinations, ensuring the system only outputs 100% accurate and verified information.

Custom Heap Memory Allocator — *C, Memory Management, Linux, Make*

Jan 2025 – Jan 2026

- Architected a custom dynamic memory allocator in C (compiled as a shared library **libheap.so**), replicating standard heap operations with strict **8-byte block alignment**.
- Engineered a **best-fit allocation algorithm** to minimize internal fragmentation, implementing block splitting to optimize space utilization for varying request sizes.
- Designed **footer-based tracking** and an **immediate coalescing** mechanism to dynamically merge adjacent free blocks, significantly reducing external memory fragmentation.
- Built a comprehensive automated test suite utilizing **Makefiles** (spanning basic allocation to advanced coalescing) and validated memory safety and zero-leak operations using **Valgrind**.

TECHNICAL SKILLS

Languages & Web: Python, Java, C, JavaScript (ES6+), Node.js, HTML/CSS, WXML/WXSS, Vue, React, SQL
Frameworks & Tools: Flask, Vite, Git, MySQL, MongoDB, Redis, Linux, Make, Postman, Figma, Slack Bolt SDK
AI & Machine Learning: RAG Pipelines, Scikit-learn, spaCy, OCR/NLP, Stable Diffusion, OpenAI API